

# Notes on practical results of Statistical Learning Theory

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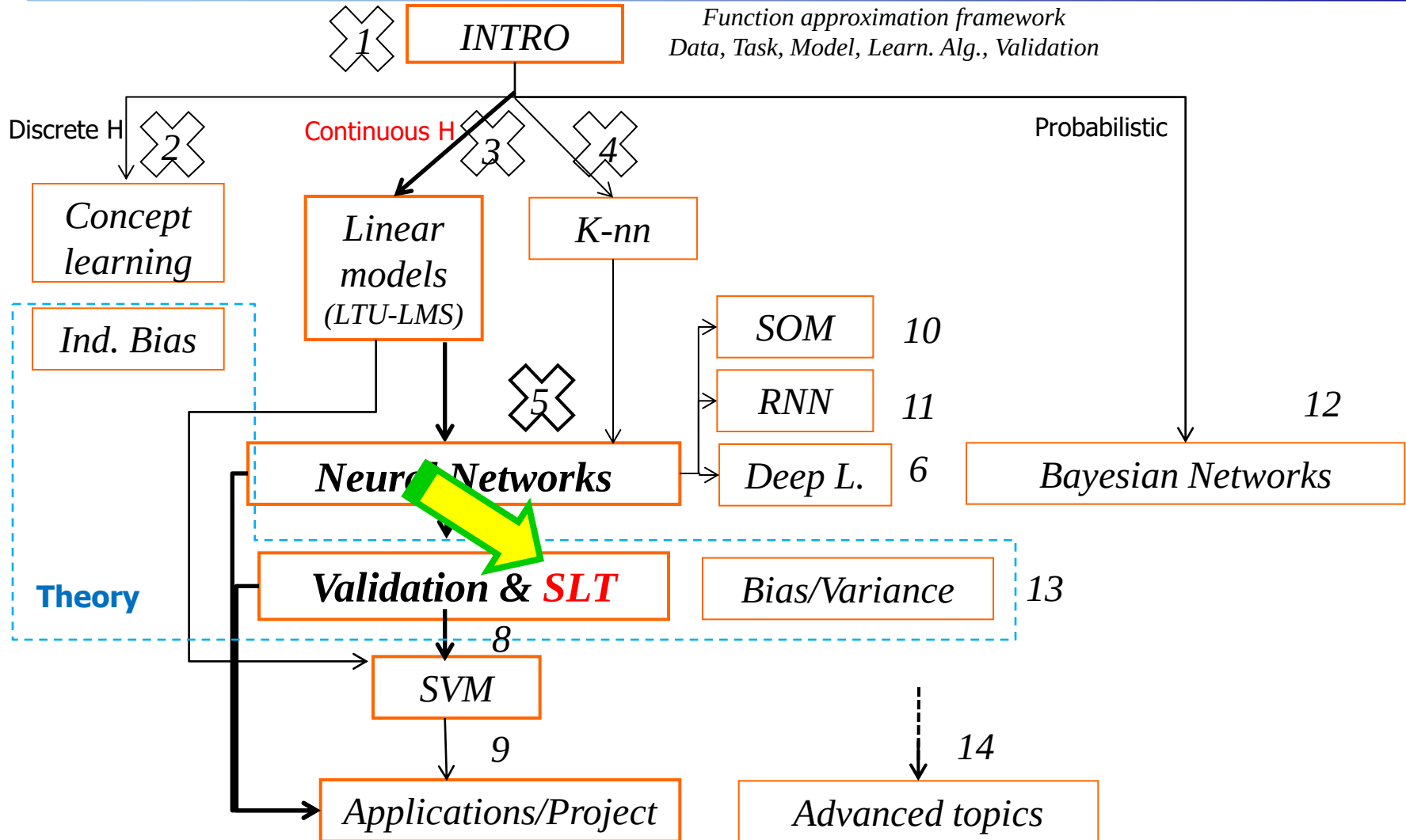
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# ML Course structure

## Where we go



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# Approximating estimation step

We can approximate the estimation of validation set error (for model selection) or test set error (for model assessment)

- Analytically:
  - AIC, BIC Akaike/Bayesian Information Criterion
    - limited to linear models (in par.)
  - MDL
  - SRM: **Structural Risk Minimization & VC-dimension**
- By resampling: direct estimation of extra-sample error via
  - cross-validation
  - bootstrap

# VC-dim

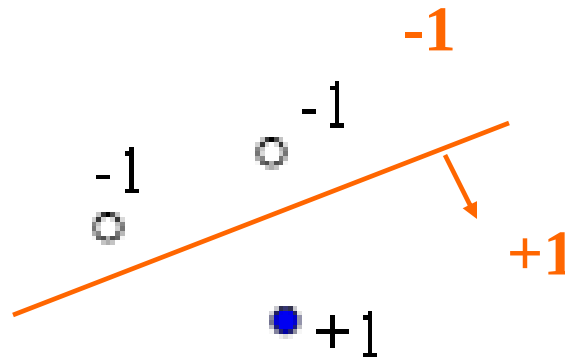
- To provide the analytical bound we need to evaluate  $H$
- A measure of complexity (capacity/expressive power/flexibility) of a class of hypotheses (also for infinite  $H$ ):
  - **V**=Vapnik  
**C**=Chervonenkis
- Not by the  $|H|$  but instead by the number of distinct instances that can be completely discriminated using  $H$
- Intuitively, restricting to the classification case:  
how much  $H$  can discriminate points ?
  - max number of points that can be correctly classified / learned without error for all the possible labelings

# Shattering (it: frammentazione)

- $X$ : instance (or input) space
- Assume  $N$  the size of (num. of instances in the) instance space (instead of  $l$  used for the num. of data, which can be a different set in concrete cases)
- $H$ : hypothesis space
- Binary classification
- There are  $2^N$  possible **dichotomies** (partitions or labeling of the  $N$  points by  $-1$  or  $+1$ )
- A particular dichotomy is **represented** in  $H$  if there exists a hypothesis  $h$  in  $H$  that realizes the dichotomy
- **Def:  $H$  shatters  $X$  iff  $H$  can represent all the possible dichotomies on  $X$  (0 errors)**
  - the points in  $X$  can be separated by an  $h$  in  $H$  in all the possible ways
  - for every possible dichotomy of  $X$ , there exists a consistent hypothesis  $h$  in  $H$

# Example: dichotomy and H

- 3 ( $M$ ) points in  $\mathbb{R}^2$



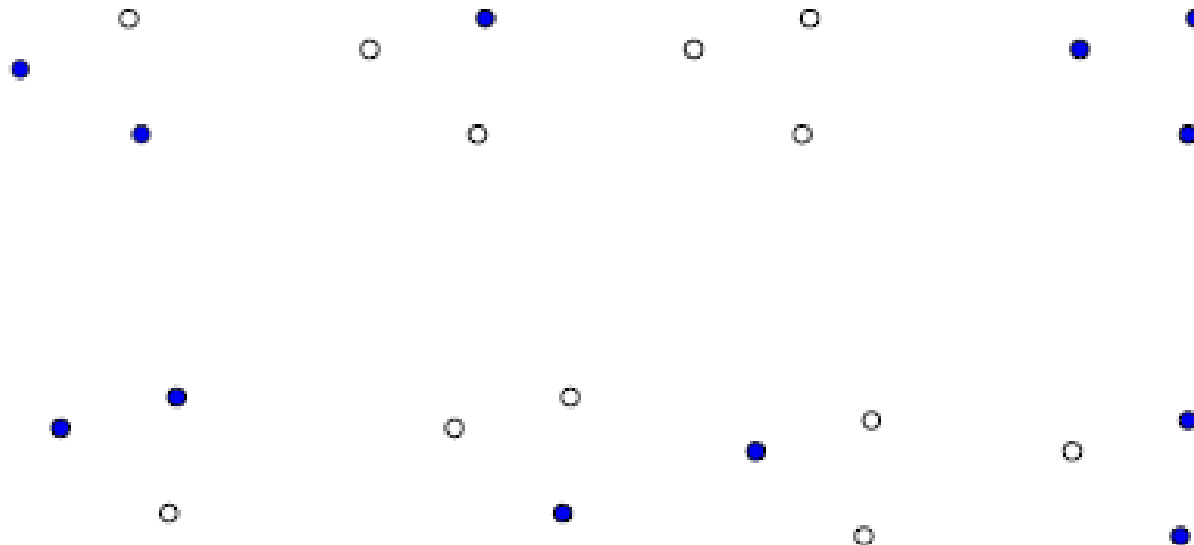
- H as set of lines  $h(\mathbf{x}) = \text{sign}(\mathbf{w}\mathbf{x} + w_0), \mathbf{x} \in \mathbb{R}^2$
- A dichotomy is a particular labeling (in  $\{-1, +1\}$ ) of the points
- This specific dichotomy can be represented in H:
  - there exist a line that correctly separates the points

# Shattering: example: dichotomies



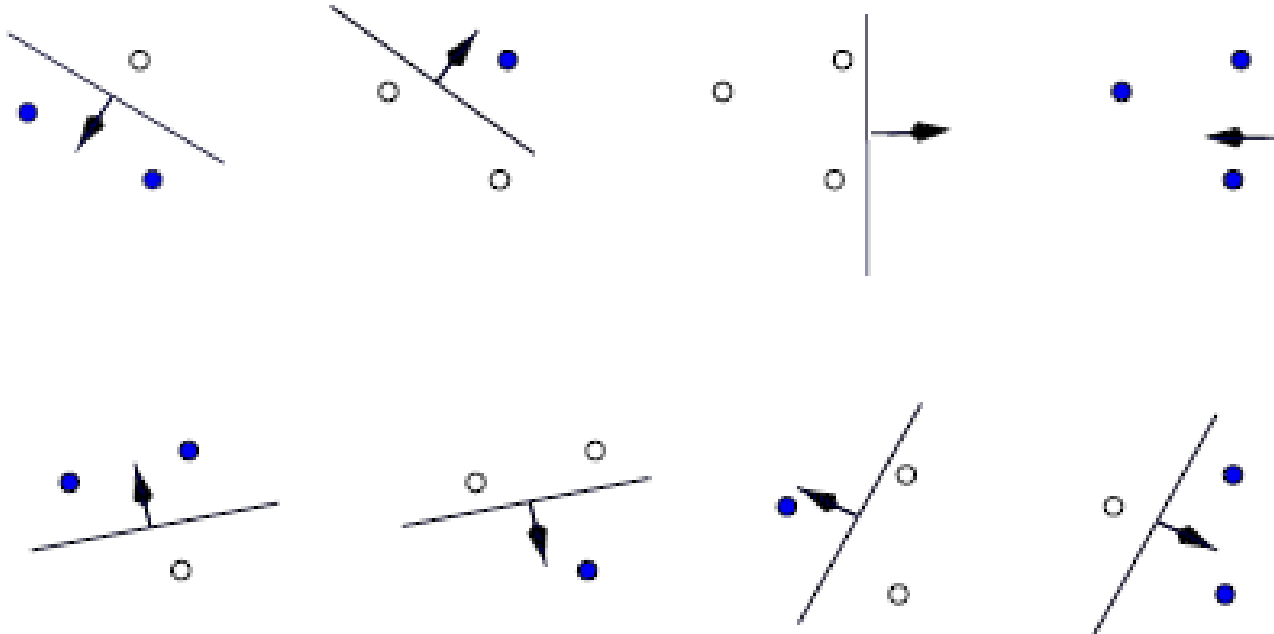
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- All the possible  $2^N$  dichotomies on 3 points in  $\mathbb{R}^2$



# Shattering: example: by linear functions

- The class of linear functions  $H$  shatters  $X$



# VC -dimension

- **Definition:** The **VC dimension** of a class of functions  $H$  is the maximum cardinality of a set (configuration) of points in  $X$  that can be shattered by  $H$
- $VC(H) = p$ 
  - $H$  *shatters* at least one set (configuration) of  $p$  points
  - $H$  *cannot shatter* any set (configuration) of  $p + 1$  points
- If arbitrarily large (but finite) sets of  $X$  can be shattered by  $H$ , then  $VC$  is infinite

# VC-dim of linear functions (1)

- $VC(H) \geq 3$  (already shown on the three points before)
- Note that however not all the possible configurations of 3 points can be shattered !

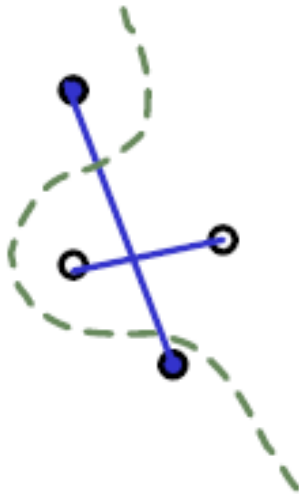
Not linearly separable



- But it is sufficient to find **one** configuration of three points which is separable for every labeling!

# VC-dim of linear functions (2)

- $VC(H) \geq 3$  (shown on the 3 points)
- $VC(H) < 4$



Can always draw six lines between pairs of four points.

Two of those lines will cross.

If we put points linked by the crossing lines in the same class they can't be linearly separated (do you remember the XOR?)

So a line can shatter 3 points but not 4

- Conclusion:  $VC(H) = 3$ ,  $H$  hyperplanes in  $\mathbb{R}^2$  (LTU)

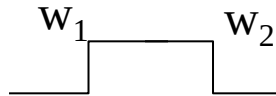
# VC-Dim: various H

- In general: The VC dimension of a class of linear (separator/decision) hyperplanes (LTU) in a  $n$ -dimensional space is  $n + 1$

*Note: upper-bound, it can be less constraining the model, we'll see how...*

**Exercise:** Try also:

- Lines in  $IR$  ( $VC(H)=2$ )
- Intervals on  $IR$  ( $VC(H)=2$ )
- Circles centered in the origin in 2D ( $f(\mathbf{x}, \mathbf{b}) = \text{sign}(\mathbf{x} \bullet \mathbf{x} - \mathbf{b})$   $VC(H)=1$ )  
( $f(\mathbf{x}, \mathbf{b}, \mathbf{k}) = \text{sign}(\mathbf{k} \mathbf{x} \bullet \mathbf{x} - \mathbf{b})$   $VC(H)=2$ )
- Circles (free radius and center) in 2D ( $VC(H)=3$ )
- Circles (fix radius, free center) in 2D ( $VC(H)=3$ )
- Can be difficult to be calculated using non-linear parameters



A circle:

$$x_1^2 + x_2^2 = r^2$$

# Number of parameters

- Is VC the number of free-parameters?
- Related but not really the same:
  - see the example in the previous slide for circles (with the same VC but different number of free-parameters)
  - and of course, we may add redundant free-parameters
  - there exist models with one parameter and infinite VC dim
    - See Haykin/Burges in bliblio
- Moreover, for nearest neighbor VC-dim is infinite

# SLT: Analytical Bound on R

Assume  $N$  = number of data ( $l$ )

$$R[h] \leq \underbrace{R_{emp}[h]}_{\text{guaranteed risk}} + \overbrace{\varepsilon(VC, N, \delta)}^{\text{VC-confidence}}$$

- *Example:* For 0/1 loss

$$\varepsilon(VC, N, \delta) = \sqrt{\frac{VC(\ln \frac{2N}{VC} + 1) - \ln \frac{\delta}{4}}{N}}$$

with probability at least  $(1 - \delta)$  for every  $VC < N$

- There are different bounds formulations according to different classes of functions, of tasks, etc.
- Interpretation: see **introduction slides on SLT** ( $\varepsilon \rightarrow 0$  increasing  $N$ ,  $\varepsilon$  grows with  $VC$ , U shape plot for  $R$  increasing the  $VC$ -dim)
- Now, this gives us a way to estimate the error on future data based only on the training error and the  $VC$ -dimension of  $\mathbf{H}$  (\*)
- And the  $VC$ -confidence can be computed prior to learning
- Note: The resulting bounds are “worst case”, because they must hold for all but  $(1-\delta)$  of the possible approximation function/ training sets.

# Remarks

Example of consequences:

- For many reasonable hypothesis classes (e.g., linear approximators) the VC dimension is linear in the number of “free-parameters” of the hypothesis.
- This shows that to learn “well”, we need a number of examples that is linear in the VC dimension (so linear in the number of parameters, in this case).

*Here's an example of a possible heuristic (on the num .of examples for the linear model) explained with the principles of the theory*

# Structural Risk Minimization

- SRM uses VC dimension as a controlling parameter for minimizing the generalization bound on  $R$
- Assuming finite VC we can define a nested *structure* of models-hypothesis spaces according to the VC-dim as in the following

$$- \mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \dots \subseteq \mathcal{H}_n$$

$$- VC(\mathcal{H}_1) \leq \dots \leq VC(\mathcal{H}_n)$$

Some example?

- NN with increasing number of Hidden units (roughly), but also the #epochs (we know that also the size of weights care and a NN is a variable-size  $H$ )
- Polynomial of increasing degree  $M$
- Increasing values for  $c$ , where  $||w|| < c$  for regularization (or controlled by  $\lambda$ )
- Increasing number of nodes in a decision tree (roughly)
- *Note: important* hyper-parameters  $\rightarrow$  related to VC-dim

# SRM for model selection

Growing VC-dim: empirical (TR) error decreases, VC-confidence increases

**Structural Risk Minimization:** find a trade-off on the bound  
(as seen in the introduction using  $l$ ):

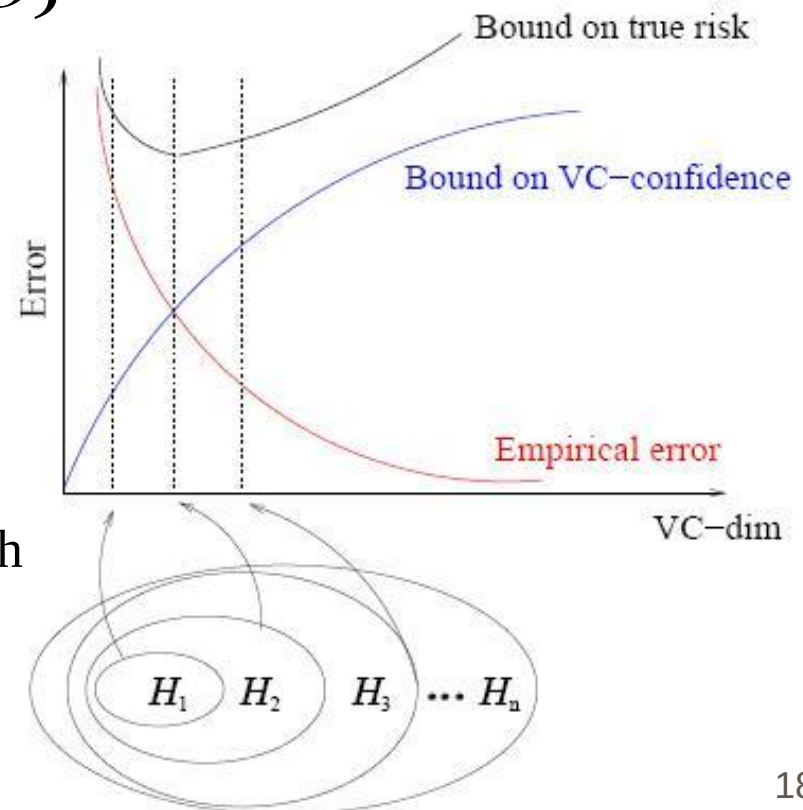
$$R \leq R_{emp} + \varepsilon(1/l, VC, 1/\delta)$$

$$- \mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \dots \subseteq \mathcal{H}_n$$

$$- VC(\mathcal{H}_1) \leq \dots \leq VC(\mathcal{H}_n)$$

e.g. NN with increasing number of  
Hidden units

**Model selection:** choose the model ( $h$ ) with  
the better bound on the true risk



# SRM: model selection

| $i$ | $H_i$ | TR err                                                                             | VC-Conf                                                                             | Probable upper bound on $R$                                                          | Choice                              |
|-----|-------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------|
| 1   | $H_1$ |   |    |   |                                     |
| 2   | $H_2$ |   |    |   |                                     |
| 3   | $H_3$ |   |    |   | <input checked="" type="checkbox"/> |
| 4   | $H_4$ |   |    |   |                                     |
| 5   | $H_5$ |   |    |   |                                     |
| 6   | $H_6$ |  |  |  |                                     |

Beware: However, the VC-Conf can result very conservative: hundreds of times larger than the empirical overfitting effect

# Use of the bound (1)

- Provide a fundamental theoretical ground for principled ML
  - Independently from specific models details or learning alg.
  - Highlight the role of complexity control
  - The optimal choice of the model complexity (on the structure) provides the minimum of the expected risk:

## **the inductive principle of SRM**

- As estimation of predictive errors? Rarely used ....
  - See next slide
- Provide a direction for new models development guided by SRM

# Use of the bound (2)

- Provide a fundamental theoretical ground for principled ML
- Estimation of predictive errors? Rarely used ....
  - It can be very loose: but it can still be useful for model selection (relative size/ranking of error is what matters)
  - + Probabilistic upper bound for all the functions  $h(x, \theta)$ : we can search over the class
  - Difficulty in calculating the VC-dim of specific class of  $H$
  - The crude upper bound (over pessimistic) may be not adequate for reliable evaluation of the value of generalization error (for model assessment)
  - Tighter bounds are under development
- Provide a direction for new model development guided by SRM
- Toward *principled approaches*: less based on empirical trial&error

# SRM Approaches

$$R \leq R_{emp} + \varepsilon(1/l, VC, 1/\delta)$$

VC-confidence

For instance, two practical approaches:

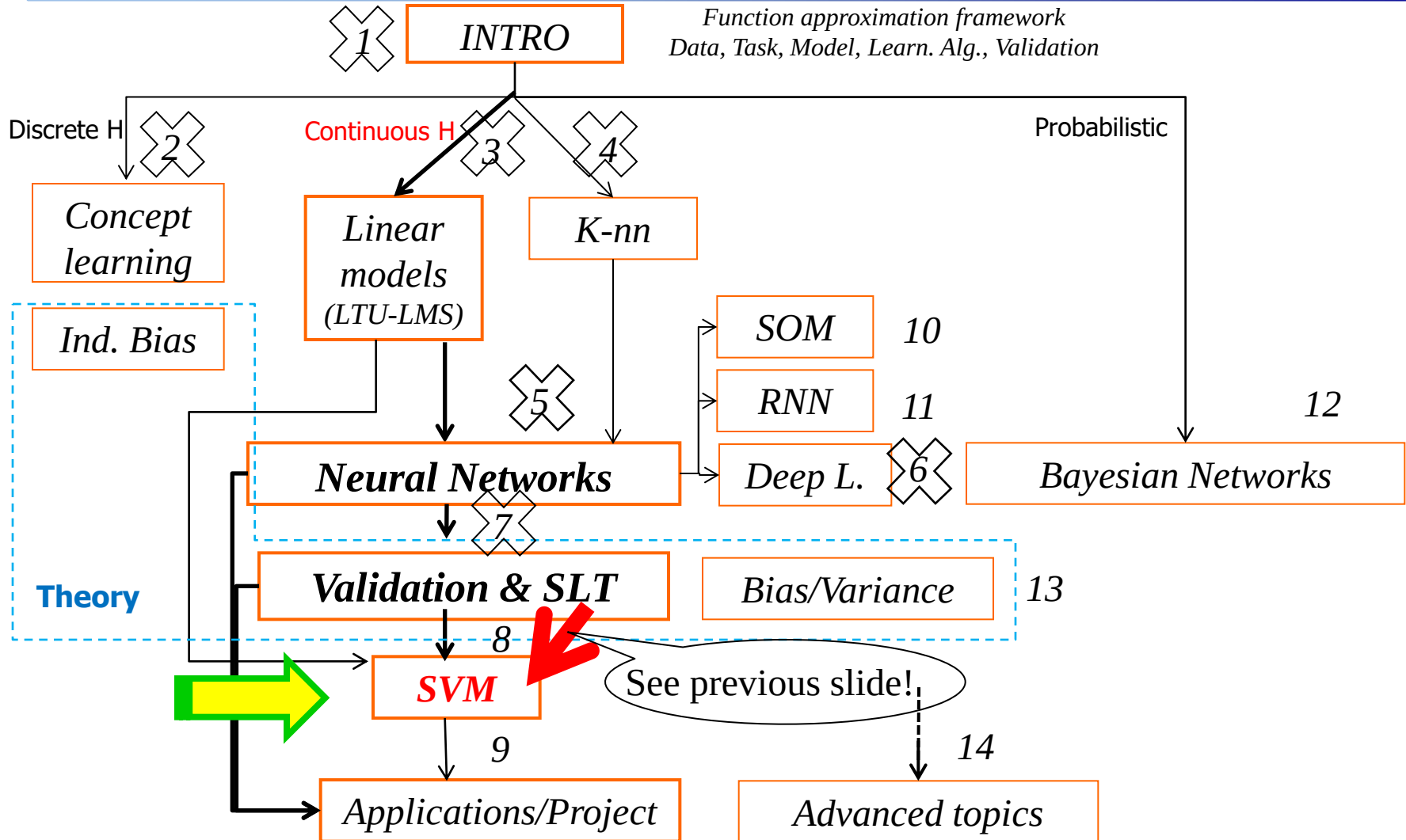
1. Choose appropriate structure/complexity, fix the model (and hence the VC-confidence), minimize TR error (e.g. can be used in NN)
  - However, regularization due to training heuristics can already introduce an *implicit* SRM implementation (early stopping, ...)
  - Or SRM by the Tikhonov regularization, where we have min loss with  $R_{emp} + \text{complexity term}$ , considering both the terms
2. Fix the TR error,  
automatically minimize the VC-confidence: how ?  
We will ... (see next lecture on SVM)

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# Bibliography

- Intro and VC-dim
  - Haykin (2<sup>nd</sup> ed.): VC dimension and SRM in chapter 2 (2.13,2.14)
  - Haykin (3<sup>rd</sup> ed.): section 4.19 (reduced view)
  - Mitchell VC-dim in Section: 7.4
  - First part of the paper: Muller, Mika, Ratsch, Tsuda, Scholkopf. An introduction to kernel-based learning algorithms. IEEE Trans. Neural Networks, 12(2):181-201, May 2001

A complete view:

- In short (but it requires not trivial math background):  
V.N. Vapnik, *An Overview of Statistical Learning Theory*,  
IEEE Transactions on Neural Networks, Vol. 10, No. 5, Sept. 1999.

Or better: See Biblio slides *after* SVM

- Vapnik, Burges, Cerkassky-Mulier, Scholkopf ...